See what matches. Prove what you checked.

Official Cognisor AI Ideathon 2026 submission. This document answers every published judging question, then the remaining submission questions in the Official Rules, so a judge can evaluate the project without a live explanation.

LANE 1 · COPY
Does this reuse a work already registered in the declared catalog?

LANE 2 · ORIGIN
What evidence exists that a generative model was involved?

LANE 3 · PROFILE
Does it resemble a specific creator's body of work?

Event
AI Ideathon 2026 · AI for Real World Impact
Theme area · Open Innovation / Business Productivity

Build
CreatorProof v0.10.0
MODEL-ACCURACY-HARDENING-2026.08.10

00 Official submission identity

Fields required by the Cognisor portal and Section 5 of the Official Rules. Paste-ready. Every link below must remain public; judges must not be asked for access.

Project title	**CreatorProof** — visual rights intelligence
Team name	CreatorProof
Member names	Ahnaf Zakaria · Mujtahidul Hasan Sami · Farhan Tanvir · Anindya Debnath Ani
Short description	Agencies, brands and marketplaces must check images before they publish, but “is this safe?” is three different questions. CreatorProof runs independent AI lanes for copy, AI-origin and creator resemblance, applies the customer’s policy as PASS / REVIEW / BLOCK, and seals the result into a receipt a third party can verify. It returns evidence, not a legal infringement finding.
Theme	AI for Real World Impact · Open Innovation, with a Business Productivity wedge
Target users	Creative agencies (beachhead), brand studios, marketplaces, publishers; creators as the free supply side
Project URL	https://creatorproof.uaenorth.cloudapp.azure.com/ · Artist `/artist` · Checker `/user`
Source code	https://github.com/AhnafZ778/CreatorProof
Demo video URL	Submitted in the portal Demo Video field (public, playable without sign-in)
Stage	Working end-to-end prototype: FastAPI evidence API + Next.js portals. Not a mockup.

PORTAL SHORT DESCRIPTION (COPY THIS; WELL OVER 40 CHARACTERS)

Publishers have to decide whether an image is safe to release. That is really three questions — reuse, AI origin, and style resemblance — and today’s tools answer them with one unverifiable number. CreatorProof keeps the questions separate, uses AI to nominate and classical geometry to verify, then records the decision as a signed receipt.

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02 Problem

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01 Judging criteria

Official Rules, Section 9. Each published question is answered in full, with evidence, and with the gaps stated rather than hidden. Weights are the organiser's.

25%	20%	20%	20%	15%
Problem fit and real-world impact	Innovation and idea quality	Technical implementation	Product potential and scalability	Demo and presentation

1. Problem fit and real-world impact

WEIGHT 25%

QUESTION 1.1

Is the problem clearly defined?

Yes. The problem is not "AI art" or "copyright on the internet." It is a pre-publication decision that organisations already have to make, and that they currently make with the wrong unit of analysis.

Before a brand, agency, marketplace or publisher presses publish, someone must answer **three different questions** about the same file:

- **Reuse.** Does this image contain a work that is already registered in the sources we are allowed to search?
- **Origin.** What evidence exists that a generative model was involved in producing it?
- **Resemblance.** Does it resemble a specific living creator's body of work, even if no copy of a stored file is present?

The market sells one number for all three — "87% AI", "93% similar", "match found." Each collapse causes a different harm: origin-as-guilt, similarity-as-infringement, style-as-authorship. A fourth failure is quieter: when a tool searches an incomplete database and returns "no matches," users hear "this image is clean." That is a definition of the problem, not a slogan: *publishers need a structured record of what was checked, what was found, and what was decided — not a verdict dressed as a percentage.*

QUESTION 1.2

Is the problem meaningful for the intended users?

Yes, for people who already carry the cost. A single disputed campaign image can exceed a year of software spend. Marketplace reviewers process takedowns with no shared evidence packet and no consistent standard between shifts. Independent illustrators have no practical way to register a work so that the organisations most likely to reuse it actually consult that registration.

INTENDED USER	WHY IT IS MEANINGFUL TO THEM
Creative agency (beachhead)	Client contracts now demand originality warranties and AI disclosure. The agency cannot currently attach a defensible check to the invoice.
Brand in-house studio	Dozens of agencies and freelancers feed the same brand. There is no consistent pre-publication standard and no audit trail six months later.
Marketplace / UGC platform	Human review queues do not scale. Appeals need structured evidence, not a moderator's memory.
Publisher / newsroom	Synthetic-media disclosure is moving from etiquette into policy. An editor's gut is not a record.
Independent creator	Not the paying customer. They need a timestamped registration that is actually searched when a brand checks an asset — funded by the checking side.

A Bangladeshi studio delivering to US/UK/EU clients is a concrete local instance of the same problem: they can lose a Western contract because they cannot *show* they checked, not because they stole anything.

QUESTION 1.3

Does the solution address a genuine real-world need?

Yes. The need already exists as unpaid labour and residual legal risk. People reverse-image search, squint, and hope. That workflow fails on the cases that matter: a mirrored crop inside a social screenshot; a file with no Content Credentials; a style-mimic that copies no stored pixels.

CreatorProof addresses the need that is actually in the job description — **produce a check you can stand behind** — by:

- searching a *declared* catalog and stating coverage before any "no match" language is allowed;
- keeping copy, origin and profile as separate evidence, never averaged into an infringement score;
- turning findings into the customer's own PASS / REVIEW / BLOCK policy, with reason codes;
- sealing the packet so a counterparty can verify it without trusting our database.

It does **not** pretend to replace a lawyer, crawl the open internet, or certify that a work is human-made. Those are not the customer's operational need; a dated, scoped, inspectable record is.

QUESTION 1.4

Could the solution create measurable value or impact?

Yes — in hours displaced, incidents avoided, and disputes that have a file instead of a Slack thread. We model the beachhead (a 20–200 person agency, ~500 assets/month) as:

VALUE LEVER	HOW IT IS MEASURED	MODELLED ANNUAL
Manual review labour	~6 minutes/asset displaced from reverse-image search	$16,800
Late pulls / rework	Campaigns withdrawn after a problem surfaces post-publish	$18,000
Claim expected value	Low-probability, high-cost disputes	$4,500
CreatorProof Studio tier	Platform + included scan/catalog volume	$7,200
Modelled net	Before counting retained-client value, which is the closer	**$32,100**

Honest gap: these figures are reasoned unit economics, not pilot measurements or letters of intent. What the prototype already makes measurable, today, is operational: scans completed, coverage state, match/review/no-match-in-checked-sources, policy action, packet hash, and (when configured) an on-chain commitment. Impact after adoption is “fewer blind publishes and a packet you can hand a client.” Impact we will not claim until a customer corpus exists: a published false-positive rate on that corpus.

2. Innovation and idea quality

WEIGHT 20%

QUESTION 2.1

Is the idea creative?

The creative move is not a new neural net. It is refusing the product category the market is in — *AI that issues a verdict* — and building the category organisations actually need: *AI that produces evidence, with a proof that the evidence existed.*

Four design choices are unusual in combination:

- **Lane separation enforced in code.** Copy, origin and profile have different data, failure modes and legal meanings. They are never fused into an “infringement percentage.”
- **Coverage before verdict.** `NO_MATCH_IN_CHECKED_SOURCES` is impossible unless declared catalog coverage is `COMPLETE`. Incomplete searches cannot impersonate clearance.
- **AI nominates; geometry verifies.** A learned descriptor may propose a candidate. A match requires independent physical alignment and aligned-pixel corroboration.
- **Joint witnessing.** A second organisation can co-attest the same evidence commitment on chain, so a decision is not a single vendor’s word.

The product’s most distinctive behaviour is abstention: it will not emit its strongest origin claim until independently verified, calibrated evidence families agree — and today it says they do not. Shipping a system that visibly refuses to overclaim is the idea.

QUESTION 2.2

Does the solution provide a meaningful improvement over existing approaches?

Existing tools optimise for one of the three questions and imply the other two.

APPROACH TODAY	WHAT IT ACTUALLY ANSWERS	WHAT USERS WRONGLY HEAR
Reverse image search	Near-duplicate pages the crawler has seen	“Not infringing” if nothing famous appears
AI-image detectors	A score under a specific training distribution	“This is / is not AI,” as a fact
Style / “who painted this” apps	Nearest embedding in a scraped set	Authorship, and therefore copying
Manual legal review	A real opinion, slowly, expensively	Unavailable at upload time

CreatorProof’s improvement is structural: **scoped search, separated questions, policy the customer owns, and a packet a third party can hash.** A PASS means “nothing against your policy in the sources we finished searching,” which is a weaker sentence and a more useful one. We also recover transformations reverse-image search often misses — mirror, crop, re-encode, screenshot chrome — because the copy lane is built for reposts, not for web-index hits.

What we do not claim as an improvement: we do not search the open internet. A tool that claims to have checked everywhere produces a “no match” that means nothing. Declared catalogs are the honest scope.

QUESTION 2.3

Is AI being used purposefully rather than simply added to the project?

Yes. Every question the product answers is a model question. The architecture exists to *constrain* what those models are allowed to conclude. That is the opposite of a chatbot bolted onto a dashboard.

Where AI is load-bearing:

- SSCD retrieves likely copies from the catalog (including regional crops so a collage still nominates).
- Community Forensics (and optional Sightengine) supply origin signal; C2PA is checked first because a signed manifest beats any detector.
- CSD embeddings support creator-profile resemblance, with a content control so “same subject” is not mistaken for “same hand.”
- Tesseract reads visible generator watermarks as a forgeable review cue, never as provenance.

Where AI is deliberately not allowed to finish the job:

- A neural score cannot declare a copy. SIFT/ORB + USAC/MAGSAC homography and aligned structure must agree.
- Origin scores are labelled `ORIGINAL_PRESERVING_QUALITY_WEIGHTED_ENSEMBLE_NOT_PROBABILITY`. They are not percentages of truth.
- The profile lane cannot escalate policy on its own. Resemblance is not copying.
- Optional OpenRouter text only explains structured metrics. It cannot change retrieval, match status, or policy.

Purpose is also governance: SHA-256-pinned model bundles, an evidence-family registry so two clones of one detector cannot fake independence, a qualification ladder that self-reports `RUNTIME_READY` rather than “production,” and `automatic_promotion_allowed: false`. AI is used to see; rules are used to decide what seeing is allowed to mean.

3. Technical implementation

WEIGHT 20%

QUESTION 3.1

Has the team developed a functional prototype or convincing technical concept?

A functional prototype, not a concept deck. CreatorProof v0.10.0 is a running system:

- **API.** FastAPI 0.10.0 — catalog, scans, policy, review, proof, network co-attestation, webhooks, usage meters.
- **Web.** Next.js 16.3 / React 19 — marketing landing, Artist portal, User (checker) portal, Evidence Microscope, proof panel.
- **Models on the copy/origin/profile path** load from a pinned bundle (SSCD, Community Forensics, optional CSD). None of those three is a UI placeholder.
- **Proof.** Canonical SHA-256 packet, Ed25519 / COSE Sign1, RFC 6962-style Merkle log, EAS attestation pipeline on Base Sepolia when configured.

Public interface: creatorproof.uaenorth.cloudapp.azure.com. Source: github.com/AhnafZ778/CreatorProof. End-to-end scans execute against the FastAPI backend on the same host. The Azure app is the real product UI; detection is not simulated in the browser.

QUESTION 3.2

Does the AI functionality work as described?

Yes, within the scope we state. We measured the copy lane on a harness we wrote (`scripts/copy_detection_bench.py`): 14 source images × 26 real-world transformations = 364 positive cases, plus 115 unrelated-image negatives.

98.08%	99.45%	0	14/14
positives returned `MATCH_FOUND`	flagged match or review	false matches on 115 negatives	horizontal mirrors recovered after a dedicated retry path

Residual failures are named: two misses, both a flat repetitive graphic under aggressive crop or small rotation — SIFT cannot lock distinctive keypoints. We did not loosen thresholds to erase those misses, because a false match is worse than a review referral.

Origin works as described, including the uncomfortable part: missing C2PA is `NOT_PRESENT` (unknown, not human); the strongest origin class is suppressed until calibrated independent families exist; we currently have one verified family, so the system abstains from that class. Profile resemblance ranks with CSLS density correction and stays advisory.

These numbers are our pipeline on our transformation set, not a public leaderboard. That is disclosed in the product, not only here.

QUESTION 3.3

Is the technical approach logical and feasible?

Logical because each stage uses the tool that is good at that stage, and feasible because every stage already runs.

01 · Retrieve	02 · Verify	03 · Decide	04 · Prove
Learned	**Classical**	**Rules**	**Crypto**
SSCD (and pHash fallback) nominates catalog works, including regional crops for collages.	SIFT/ORB, ratio test, USAC/MAGSAC homography, mirror retry, aligned luminance/gradient/SSIM consensus.	Typed fusion + versioned customer policy. Fusion is not a second neural net, because a customer must be able to justify it.	Deterministic packet hash, issuer signature, Merkle inclusion, optional 32-byte EAS attest. Artwork never goes on chain.

Feasibility constraints are designed in: image size caps; catalog completeness bounds (above the reference limit, a scan cannot claim complete no-match); crash-safe persist-before-broadcast for chain writes; idempotency bound to bytes + catalog + intended use. The approach scales first as an API with remote embeddings (the Render profile), then as GPU workers — not as a laptop demo that cannot leave the laptop.

QUESTION 3.4

Can the team clearly explain how the system works?

Yes. One sentence, then the moving parts.

> THE SYSTEM IN ONE SENTENCE
>
> You declare a catalog and an intended use, we run three independent evidence lanes over the image, we apply *your* policy, and we hand you a signed packet whose hash can be checked without trusting us.

A creator registers a reference in the Artist portal; an AI-origin gate can refuse enrollment and returns the score and threshold so the refusal is contestable. A checker drops a candidate in the User portal, sees catalog size *before* the scan (so an empty catalog cannot look like clearance), watches stage progress, then reads: one bottom line, three lane cards, an Evidence Microscope with actual keypoint correspondences, and a proof status. Policy is a versioned document with a dry-run endpoint. Proof is four independently checkable layers: canonical hash, signature, transparency log, public anchor.

This PDF, the README, and the live UI use the same vocabulary on purpose: match status, policy action, coverage, not “infringement %.”

4. Product potential and scalability

WEIGHT 20%

QUESTION 4.1

Could the solution develop into a usable product, startup, or scalable initiative?

Yes. The prototype is already shaped as a product, not as a notebook: multi-tenant API keys, PostgreSQL row-level security, idempotent scans, append-only usage meters (scans, protected assets, storage, GPU seconds, anchors), signed webhooks, first-party Python and TypeScript SDKs, versioned policies, review cases.

The startup shape is a B2B API with two portals. Revenue is a three-part tariff already instrumented in code: platform fee (audit continuity), catalog fee (network asset), metered scans (value alignment). Studio / Growth / Platform list prices are $299 / $1,200 / $4,500 per month, with forensic and anchor overages.

Scalability is catalog-shaped: each registered work makes every future scan more useful to every customer who shares or consults that catalog. Co-attestation makes switching costly once two organisations have jointly witnessed a decision. Those are product properties, not slide-deck network effects.

QUESTION 4.2

Is there a clearly identifiable target user or customer?

Yes. We do not start with “everyone who uses images.”

Beachhead customer	Creative agency or production studio, 20–200 people, running paid campaigns
Economic buyer	Head of production or operations director — one signature, no procurement theatre
Champion	Producer / traffic manager who currently performs the check by hand
Trigger	A near-miss, a new MSA originality clause, a client AI-disclosure questionnaire, or an actual claim
Disqualifiers	No contractual originality terms; very low volume; expects a legal determination or a web-wide crawl
Not the buyer	Consumers wanting a free “is this AI?” button; independent artists (supply side, free registration)

Expansion path is built into the artefact: the agency’s proof packet is seen by every client it serves, which is distribution into brands without a second sales motion.

QUESTION 4.3

Can the solution grow beyond the hackathon?

Yes. The next product steps are customer-corpus calibration, a first agency pilot, live Base Sepolia (or mainnet L2) anchoring in the default demo path, and catalog network effects between agencies that already share talent.

Technical growth is already sequenced in the repo: remote embedding workers so the API does not ship 2+ GB of weights in every process; S3-compatible media; review queues for the REVIEW bucket; SSO and private deployment for enterprise. None of those require rewriting the evidence model. The lanes, coverage rule, and packet schema are the durable core.

What does not automatically grow: accuracy on a customer's actual archive. That takes labelled pairs and a human promotion record. The qualification ladder exists so we cannot paper over that with a version bump.

QUESTION 4.4

Is there potential for adoption, commercialization, partnership, or future funding?

Adoption. The buyer already has budget under "production risk" and "client compliance." Integration is an API plus two portals, or an SDK in an existing DAM / traffic tool. Switching cost rises with retained audit history and co-signed packets.

Commercialization. Metered B2B SaaS with a free creator registration surface. Meters already exist as append-only records; billing is an invoicing layer, not a new architecture.

Partnership. Natural counterparts are stock/licensing platforms (intake gate + co-attestation of licence decisions), marketplace trust-and-safety, C2PA / Content Credentials implementers, and agency holding groups that want one standard across shops.

Funding. The case is infrastructure with a wedge, not a consumer app: acute pain, small buying committee, expanding catalog as a moat. We do not claim a round in progress. We do claim a product that is fundable once a pilot produces a measured review-hour delta and a zero-surprise false-match record on that customer's corpus.

5. Demo and presentation

WEIGHT 15%

QUESTION 5.1

Is the problem communicated clearly?

Yes, as three questions a non-technical buyer already recognises, not as a research abstract. The landing line is the problem: *See what matches. Prove what you checked.* The supporting line states the three questions, the policy outcomes, and the legal boundary in one paragraph — the same paragraph used on the live site.

In the video and the four-minute live pitch we open with a concrete scene (a mirrored, cropped repost of a registered work) rather than a market-size slide, so the problem is visible before it is named.

QUESTION 5.2

Is the proposed solution easy to understand?

The UI is built so a reviewer who has never read this PDF can still operate it: one bottom line, then three question cards (Was AI used? Does it reuse a work? Does it resemble a creator?), then optional forensic detail. Colour encodes lane identity, not guilt. PASS / REVIEW / BLOCK are the only decision colours.

The conceptual load is four words: **submit → evidence → policy → proof.** We repeatedly say what it is not, because a misunderstood "AI copyright judge" would be easy to understand and wrong. Clarity here includes the boundary.

QUESTION 5.3

Does the demonstration effectively show the project's main functionality?

The demo is the real pipeline, not slides of the pipeline. Main functionality shown:

1. **Register** a human-claimed work in `/artist`. An obviously generative file is refused at the origin gate, with score and threshold visible.
2. **Scan** a hard derivative in `/user`: mirrored, cropped, re-encoded JPEG — not an identical file.
3. **Watch** stages run; lanes resolve independently; copy reports a match.
4. **Inspect** the Evidence Microscope: correspondence lines on pixels, aligned region, mirrored label if applicable.
5. **Read** coverage (what was searched) and the signed receipt / proof status.

That sequence is the product. A recorded backup of the same path exists for the Final Pitch Round if live networking fails, as the Official Rules require.

QUESTION 5.4

Can the team communicate its value within the allocated presentation time?

Yes. Official Rules allocate a **3-minute video** for all teams and, for finalists, a **4-minute live pitch plus 2–3 minutes of questions**. We treat those as hard clocks, not as "about four minutes."

SLOT	WHAT WE SPEND IT ON
Video · 3:00	Problem (who hurts) → solution (three lanes + receipt) → live prototype → impact → one-sentence future. No architecture tour.
Live · 4:00	~45s problem, ~45s architecture-in-plain-language, ~2:00 live register+scan+microscope, ~30s impact/ask, ~20s boundary ("not a legal finding").
Q&A · 2–3:00	Prepared on: false-match posture, why we don't crawl the web, origin abstention, who pays, what is on chain (32 bytes), what was built during the event.

Value, in the time we have: *a check you can show a client, on the cases reverse-image search misses, without pretending to be a court.*

02 Problem

Official Rules — Submission PDF item 1. Also the video-pitch problem beat.

What real-world problem are you solving?

Commercial image pipelines move faster than they can account for. A file can be generated, commissioned, scraped, mirrored, screenshot and published under a brand name in an afternoon. The organisation still has to decide whether publishing it is defensible, and today that decision is unstructured: a person, a reverse-image tab, and no artefact that survives contact with a client, a moderator appeal, or a lawyer six months later.

Generative models made origin visually ambiguous. Style mimicry became a one-prompt operation. Content Credentials (C2PA) shipped in cameras and editors, which created a standard to build on — and a new failure mode when the manifest is absent. Disclosure obligations moved into contracts and platform policy. The problem is the missing **record** at the moment of publication.

Who experiences this problem?

Production and legal-ops people inside agencies and brands; trust-and-safety reviewers on marketplaces; picture desks in newsrooms; and the illustrators whose work is the catalog those organisations ought to be searching. Neither the brand manager nor the artist is the villain. The villain is a supply chain with no receipts.

Why does it matter?

Because the cost of a wrong publish is asymmetric. A false accusation damages a living artist. A missed reuse damages a rights-holder and the publisher's client relationship. A confident "no match" on an incomplete search manufactures liability. Creative work is a large commercial surface with none of the batch records that every other regulated industry takes for granted. It matters now because the old heuristic — "I can tell by looking" — stopped working.

03 Solution

Official Rules — Submission PDF item 2. Also the video-pitch solution beat.

What did you build?

CreatorProof is a pre-publication creative-rights evidence API and two portals. It runs three independent AI lanes over a candidate image, keeps their findings separate, applies a versioned customer policy to produce PASS, REVIEW or BLOCK, and seals the result into a signed evidence packet. Optionally, a 32-byte commitment is attested on a public chain. The artwork never is.

How does the solution address the problem?

STORED-WORK COPY

Learned retrieval nominates; geometry and aligned pixels verify. Answers which registered work is present, and where in the frame.

AI ORIGIN

C2PA first, then a detector ensemble under stress views, then OCR of visible labels. Answers what evidence of generative involvement exists — and how strong it is.

CREATOR PROFILE

Style embeddings against consent-backed profiles, with a content control. Advisory only. Answers resemblance, never authorship.

Coverage is reported before any clearance language. Policy belongs to the customer, so two tenants can treat the same findings differently and both have an audit trail. Proof makes the trail portable: a counterparty hashes the packet, checks the signature, and — if anchored — checks a chain nobody in the deal controls.

04 AI implementation

Official Rules — Submission PDF item 3: models, APIs, agents, data, automation, workflows, limitations.

Models

MODEL	ARCHITECTURE	RUNS	ROLE
SSCD `disc_mixup`	ResNet-50 family, 512-d	Local · ~95 MB	Copy-detection retrieval. Meta AI Research; trained for copies, not generic similarity.
CSD ViT-L	ViT-L, 768-d style	Local · ~2.3 GB	Creator-profile resemblance. Optional; transparent low-level style fallback if absent.
Community Forensics	ViT-S/16 @ 384px	Local · ~84 MB	Synthetic-image detector across many generator families. Official eval transform (short side 440, center crop 384).
Sightengine `genai`	Vendor API	Cloud · optional	Primary origin opinion when configured. Local detectors are fallback, not a second vote to shop for a better score.
Tesseract OCR 5.3.4	LSTM text	Local CLI	Visible “AI generated” labels. Review signal; never trusted provenance.
`c2patool` 0.27.7	C2PA implementation	Local CLI	Signed Content Credentials. Pinned binary digest. Absence is unknown, not human.

Neural artefacts are pinned by SHA-256 in the model-bundle manifest and validated at startup. A silently swapped weight file is a boot failure, not a mystery later.

APIs and agents

There is no autonomous agent that publishes, sues, or “decides infringement.” The product is a request/response evidence service: `POST /v1/scans` returns a scan id; detection runs asynchronously; the

UI polls stages. Optional OpenRouter is an explainer over structured metrics only. Optional remote embedding HTTP services supply SSCD/CSD vectors in the lean Render deployment so the API container does not load PyTorch.

Data

Tenant catalogs of registered reference works (fingerprints, cached descriptors, claim state). Consent-backed creator-profile manifests. Candidate bytes retained on a short window. PostgreSQL with RLS for tenant isolation; SQLite for local development; private object storage for media. Evaluation data for the copy benchmark is 14 sources × 26 transforms, produced by us, not a scraped public leaderboard. No personal data is written on chain — only a 32-byte hash.

Automation and AI workflows

1. Intake: image + catalog + intended use + idempotency key.
2. Copy workflow: regional SSCD nomination → geometric verify (mirror retry if needed) → aligned structure → rule-based fusion.
3. Origin workflow: C2PA → optional Sightengine on original bytes → local detector under ten stress views → OCR → family-aware fusion with abstention.
4. Profile workflow: CSD (or transparent fallback) → CSLS ranking → content confound control → advisory card only.
5. Policy workflow: versioned rules consume typed findings; dry-run available; default informational origin cannot independently flip policy.
6. Proof workflow: canonicalise → sign → Merkle append → optional EAS attest / counterparty co-attest.

Limitations (AI)

- There is no universal AI-image detector. We refuse to label fused scores as probabilities.
- High-authority origin outcomes require calibrated independent evidence families. Calibration on a customer domain is not present; the bundle is `RUNTIME_READY`.
- Copy accuracy is measured on our transformation harness, not on a publisher's archive.
- Style resemblance is not attribution and cannot escalate policy alone.
- Catalog search is declared-scope only. We do not crawl the web.
- AI coding assistants (Cursor and frontier models) were used to implement and test. Thresholds, scope boundaries and honesty constraints were human decisions.

05 Product and technical approach

Official Rules — Submission PDF item 4. Technologies and how the prototype works.

How the prototype works

The browser never holds an API key. Next.js route handlers talk to FastAPI. A scan is accepted immediately, processed in the background, and recovered by id. Reusing an idempotency key with different bytes, catalog or intended use returns `409 IDEMPOTENCY_PAYLOAD_MISMATCH` rather than a silent stale result. Reviewers see coverage, a bottom line, three lanes, then forensic overlays only when they ask. Proof status distinguishes a local Merkle receipt from a mined EAS anchor and will not conflate them in the UI.

Technologies used

LAYER	TECHNOLOGY
API	Python 3.12 · FastAPI · Pydantic v2 · SQLAlchemy · Alembic · uv lockfile
Inference	PyTorch / TorchScript · timm · safetensors · OpenCV · Pillow · NumPy
Data	PostgreSQL + RLS · SQLite (dev) · S3-compatible object storage
Web	Next.js 16.3 · React 19 · TypeScript · tokenised CSS · self-hosted fonts
Chain / crypto	EAS · Base Sepolia · web3.py · Ed25519 · COSE Sign1 · RFC 8785 JCS · RFC 6962 Merkle · EIP-712
Provenance	C2PA via pinned c2patool · Tesseract OCR
Operations	Prometheus-style metrics · stage timelines · signed webhooks · usage meters
Deploy (demo)	Azure VM creatorproof.uaenorth.cloudapp.azure.com · GitHub Actions hygiene
Build-time AI tools	Cursor · frontier LLMs for implementation and critique · ElevenLabs for pitch narration

06 User experience

Official Rules — Submission PDF item 5. How someone actually uses the product.

Creator — Artist portal `/artist`

1. Opens the Artist portal. Sees library count, whether a creator profile has enough works (≥3), and asserted claims waiting for corroboration.
2. Drops in a reference image, names the work, states a claimant and catalog.
3. The origin lane runs as an admission gate. If the fused score exceeds the configured threshold, registration is refused and the refusal shows score, threshold and summary — contestable, not a silent 403.
4. On success they hold a registration receipt. The work is now searchable by checkers using that catalog.

Publisher / reviewer — User portal `/user`

1. Sees how many works the named catalog currently holds *before* scanning, so an empty catalog cannot look like a clean bill of health.
2. Drops a candidate (or uses a prepared demo scenario). Stages stream: intake, retrieval, geometry, origin, policy, proof.
3. Reads the bottom line first (policy action + match status), then the three independent question cards.
4. Opens the Evidence Microscope only if they need to disagree with the machine on the same pixels the machine used: inliers, transfer error, aligned region. Connection lines stay quiet until hover unless the reviewer asks for all of them.
5. Opens proof: packet hash, signature, log inclusion, anchor status. Optionally a counterparty co-attests.

The experience is built for a producer under time pressure, not for a researcher. Forensic density is progressive disclosure. The product copy on screen matches the legal boundary in this PDF: evidence and policy, not infringement.

07 Responsible AI

Official Rules — Submission PDF item 6: privacy, security, bias, transparency, safety, human oversight.

PRIVACY

Catalog images stay in tenant storage. Chain writes are a hash, never the artwork, never a name. Creator profiles are consent-backed manifests, not a scrape of the internet.

SECURITY

API keys map to tenant + role. RLS denies unbound queries. Browser holds no secrets. Idempotency and crash-safe chain persistence prevent duplicate or lost writes. Credentials are not in the repository.

BIAS AND ERROR POSTURE

False matches are treated as more damaging than misses. Profile hubness is corrected with CSLS. Tiny catalogs cannot manufacture high-confidence creator attribution. Multiplicity correction applies when many profiles are compared.

TRANSPARENCY

Every result carries coverage, reason codes, model-bundle identity, and limitations. Scores are not labelled probabilities. Local Merkle is not labelled as blockchain. Origin absence is unknown.

SAFETY

The system cannot authorise a rights-based pass on uncorroborated claims. Informational origin cannot independently change policy in the default mode. Incomplete scope routes to REVIEW, never to silent PASS.

HUMAN OVERSIGHT

REVIEW is a first-class outcome. The microscope exists so a human can overrule with shared evidence. Model promotion requires a human record; automatic promotion is disabled. Co-attestation is a human organisation binding its key.

08 Current prototype

Official Rules — Submission PDF item 7. What currently works vs what we plan later.

WHAT CURRENTLY WORKS

- End-to-end register → scan → lanes → policy → packet
- SSCD retrieval, geometry, mirror recovery, aligned structure
- C2PA, local origin detector, optional Sightengine, OCR labels
- Creator-profile resemblance with content control
- Coverage states; no complete no-match on incomplete scope
- Artist origin gate with contestable refusal
- Evidence Microscope and proof panel
- Ed25519/COSE, Merkle log, EAS pipeline, EIP-712 co-attestation
- Multi-tenant keys, RLS, idempotency, meters, webhooks, SDKs
- Public web + API deploy on Azure: creatorproof.uaenorth.cloudapp.azure.com
- Internal copy-detection benchmark with named residual failures

WHAT WE PLAN TO BUILD LATER

- Domain calibration on a real customer corpus; promotion beyond `RUNTIME_READY`
- Default-path live testnet transaction in every demo environment (pipeline exists; needs funded key + schema UIDs)
- Internet-scale catalogs (today: declared catalogs with an explicit completeness bound)
- Clearance-receipt token minting in the API (Solidity exists locally; not a product feature yet)
- SSO, private-cloud enterprise deploy, indemnity/procurement pack
- Pilot-measured review-hour and false-match metrics
- Video / audio lanes (image-first by design today)
- Billing provider on top of meters that already exist

DO NOT SCORE THESE AS IF THEY ALREADY SHIPPED

Live Base Sepolia in the default developer mode, production-qualified accuracy, and customer LOIs are later. Where the UI would be tempted to imply them, it labels the weaker thing instead (local Merkle ≠ chain; not-probability ≠ percent AI; REVIEW ≠ PASS).

09 Impact and future potential

Official Rules — Submission PDF item 8. Also video-pitch impact and future beats.

Impact

Near-term impact is operational: fewer blind publishes, fewer hours of unstructured reverse image search, and a packet a studio in Dhaka can attach to a delivery for a London client. Creator-side impact is that registration is free and is actually consulted by the organisations who pay — the same incentive design that made content-ID systems work, without claiming we are YouTube.

Wider impact, if the packet becomes a habit: commercial image supply chains gain an audit primitive they currently lack, and “no match” stops meaning “we didn’t look.” That is real-world impact under the ideathon theme: measurable, situated in existing jobs, and honest about remaining legal work that still belongs to humans.

Future potential

- **Product.** Agency wedge → brand pull-through via the packet → marketplace and stock-platform partnerships.
- **Network.** Shared catalogs and co-attestation turn a vendor into a standard. Leaving means leaving jointly witnessed history.
- **Geography.** Bangladesh as a production hub serving disclosure-heavy Western buyers is a launch market, not a limit.
- **Category.** Evidence infrastructure for creative publishing, analogous to batch records in other industries — not another detector widget.

10 Video pitch answers

Official Rules, Section 7. The public demo video covers these five beats with the working prototype on screen, not slides alone.

VIDEO BEAT 1 · THE PROBLEM

What problem are you solving, and who experiences it?

Organisations that publish images cannot currently produce a defensible, scoped record of what they checked. Agencies, brands, marketplaces and publishers experience it as risk; artists experience it as reuse without leverage.

VIDEO BEAT 2 · THE SOLUTION

What have you built, and how does AI contribute?

An evidence API and two portals. AI retrieves likely copies, estimates origin signal, and ranks style resemblance. Classical geometry, customer policy, and cryptography finish the job so the output is evidence, not a vibe.

VIDEO BEAT 3 · THE PROTOTYPE

Demonstrate the main functionality.

Register a work; refuse an AI file at the gate; scan a mirrored crop; show lane separation; show correspondences on the pixels; show the receipt. That is the demo.

VIDEO BEAT 4 · THE IMPACT

What is the potential value and real-world impact?

A check that belongs in the delivery file. Modelled agency value is on the order of tens of thousands of dollars a year before counting retained clients; the deeper impact is that “we looked” becomes something you can prove.

VIDEO BEAT 5 · THE FUTURE

How could the project develop beyond the ideathon?

Calibrate on a real agency corpus, run a paid pilot, turn co-attestation into a network of rights counterparties, and keep the same lanes as the product grows into DAM and marketplace integrations.

11 Built during AI Ideathon 2026

Official Rules, Section 3. Organizers may request the repository and commit history. This is the account we will give.

The submitted core is CreatorProof v0.10.0 in github.com/AhnafZ778/CreatorProof. Public history on this repository runs 10–14 August 2026, from the v0.9.2 publish through model-accuracy hardening, the artist origin gate, multi-party co-attestation, production-shaped platform work (RLS, meters, webhooks), and the Azure demo deployment. Permitted building blocks — open-source models, C2PA, EAS, FastAPI, Next.js, and AI coding assistants — are used as the rules allow. The core solution being judged is this evidence system, not a wrap of a single third-party “detect AI” API.

We can walk a judge through what a clone of the repository actually runs: tests, the copy benchmark harness, both portals, and the proof pipeline. We will not claim a live chain transaction in environments where the EAS signer is unset; the UI already refuses to imply one.

12 Demo, links, and live pitch

Official Rules: working project URL, public permissions, 3-minute video, finalist 4+2–3 live format, recorded backup if live demo fails.

Live product	`https://creatorproof.uaenorth.cloudapp.azure.com/`
Artist portal	`/artist` — register and origin gate
Checker portal	`/user` — scan, microscope, proof
Source	`https://github.com/AhnafZ778/CreatorProof`
Demo video	Public URL on the Cognisor portal; prototype footage, not slides-only
Finalist backup	Recorded register → mirrored-crop match → microscope → receipt, for use only if live networking fails

What we will not misrepresent

Official Rules, Section 10, prohibit misrepresenting prototype functionality or AI use. This project does not determine copyright infringement, certify human origin, search the whole internet, or treat a detector score as a probability. Judges who open the UI will see those boundaries in the product, not only in this PDF.

> **IF YOU ONLY REMEMBER ONE THING**
>
> CreatorProof is infrastructure for a check that already has to happen. AI does the seeing. Policy and proof do the standing-behind. The restraint is the product.

CreatorProof v0.10.0 · AI Ideathon 2026 · Cognisor platform impacthackathons.cognisorai.com. Figures in §01 / Q3.2 were measured on the authors' hardware from `scripts/copy_detection_bench.py` and are reproducible from the repository. CreatorProof produces review evidence, not legal determinations. Team names in this PDF must match the official portal listing.